Image-Conditioned Sufficiency for Sketch-Style Multimodal Reasoning

Working Title

Image-Conditioned Sufficiency: When a Sketch May Delete a Reasoning Step.

Problem Statement

SketchThinker-R1 cuts reasoning tokens by rewarding a surface “sketch” style whose judge never sees the image or the question. The claim of no accuracy loss is shown only for already-short traces on 3B and 7B models, and it is confounded with a GPT-5 rewrite of the training text.

Proposed Solution

We replace the style label with a sufficiency reward: a step may be deleted only when the remaining trace still determines the answer given the image and the question. Brevity is applied below a difficulty threshold and withheld when a step carries visual evidence, so hard items are not shortened uniformly. The same rewriter also produces a length-preserving control, which separates distillation from compression. A counterfactual image that changes one omitted visual fact tests the deletion: if the answer changes and the sketch text does not, that step should have been kept.

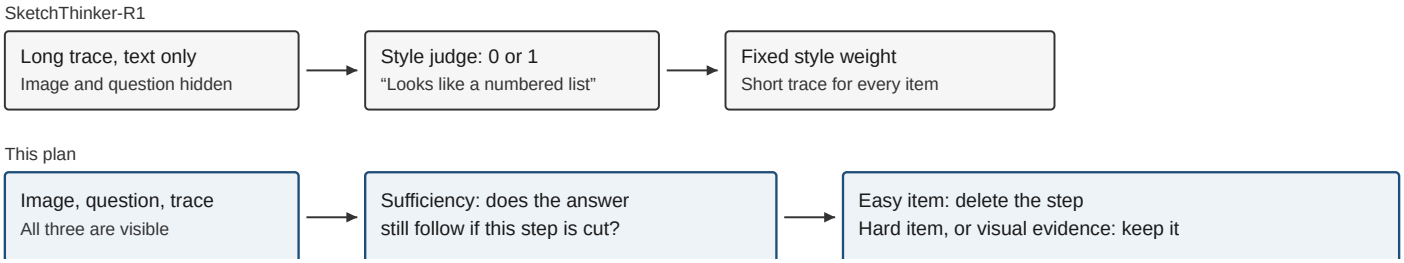
Key Research Questions

- Does text-only sketching disproportionately delete visual-evidence steps?
- At a matched token budget, does image-conditioned sufficiency recover accuracy on high-difficulty and perception-heavy items?
- After the rewriter and the training budget are matched, does any remaining gain come from sufficiency rather than from distillation?

Expected Outcome

A deletion rule tied to visual evidence and difficulty, and a measurement of whether sketch-style efficiency survives once those factors are controlled.

Figure 1. What is allowed to be deleted.



The top row is the paper’s reward. The bottom row is the proposed reward. A one-fact edit of the image is the check on the bottom row: if the answer changes while the sketch text does not, the deleted step was evidence.

Comparison

	SketchThinker-R1	This plan
What is rewarded	Sketch style: numbered and short	The remaining steps still entail the answer
What the judge sees	The thinking text only	Image, question, and trace
When the trace is short	A fixed reward weight for every item	Only below a difficulty threshold
Control for the rewriter	None. GPT-5 both shortens and may rewrite content	The same rewriter also writes a length-preserving control
Result that would refute the plan	—	Sufficiency does not beat the style reward once the budget and the rewriter match, or a one-fact image edit changes the answer while the kept trace never states that fact